

# Thermal Control Module

## Verification Test Specification

|             |                     |                |                       |
|-------------|---------------------|----------------|-----------------------|
| Document ID | TS-042              | Revision       | 1.3                   |
| Status      | Released            | Test date      | 2026-08-07            |
| Owner       | Systems Engineering | Classification | Internal test fixture |

### 1. Scope

This specification defines the bench tests for a 24 V closed-loop thermal control module. The module regulates an aluminium test plate between 20 and 80 degrees C while reporting temperature, supply current, and fault status.

### 2. System overview

The test article contains three functional blocks. Their signal flow is shown below; arrows indicate the intended reading direction from left to right.

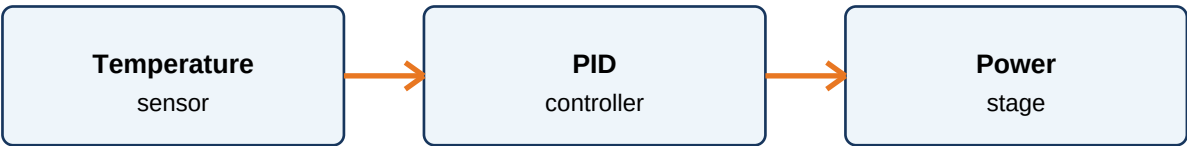


Figure 1 - Closed-loop thermal control signal path

#### Key interfaces

- Sensor input: Pt1000, four-wire connection
- Power input: 24 V DC nominal, 6 A maximum
- Data interface: isolated RS-485 at 115200 bit/s

### 3. Control model

The expected steady-state heat balance is expressed by  $Q = m * c_p * \Delta T$ . Controller efficiency is calculated as  $\eta = P_{load} / P_{input}$ .

#### Engineering note

Measurements are valid only after the plate-temperature slope remains below 0.1 degrees C per minute for five consecutive minutes.

## 4. Performance measurements

Table 1 contains the reference measurements captured at five operating points. Numeric values use SI-compatible units and a decimal point.

| Test point | Ambient (deg C) | Input (V) | Load (A) | Plate (deg C) | Efficiency (%) |
|------------|-----------------|-----------|----------|---------------|----------------|
| TP-01      | 20.1            | 24.02     | 1.20     | 30.0          | 91.2           |
| TP-02      | 20.0            | 24.01     | 2.35     | 40.0          | 92.8           |
| TP-03      | 22.4            | 23.98     | 3.60     | 55.0          | 93.5           |
| TP-04      | 25.2            | 23.96     | 4.80     | 70.0          | 92.9           |
| TP-05      | 25.0            | 23.94     | 5.50     | 80.0          | 91.7           |

Table 1 - Thermal performance at nominal supply voltage

## 5. Verification procedure

1. Inspect the wiring and confirm protective-earth continuity.
2. Apply 24 V DC with the current limit set to 6 A.
3. Command each plate-temperature setpoint in ascending order.
4. Wait for the stability condition defined in Section 3.
5. Record voltage, current, plate temperature, and fault status.

## 6. Acceptance criteria

| Requirement       | Limit          | Result                               |
|-------------------|----------------|--------------------------------------|
| Temperature error | +/- 1.0 deg C  | Pass if every point complies         |
| Input voltage     | 23.5 to 24.5 V | Pass if no dropout occurs            |
| Efficiency        | >= 90.0%       | Pass if measured after stabilization |
| Communication     | 0 frame errors | Pass over 10,000 frames              |

## 7. Expected reading order

A correct conversion reads Sections 4 through 7 in sequence, keeps each table row intact, and places this final paragraph after the acceptance table.